INDIAN JOB MARKET ANALYTICS

Data Analytics Mini-Project · Team 4

A concise analysis of job demand, fresher-friendly opportunities, work mode, experience, and advertised salary patterns in Indian job postings.

Team members

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Prepared from the submitted Jupyter notebook

1. Project overview

The project explores Indian job postings to help students and early-career job seekers understand role demand, location patterns, experience expectations, work modes, fresher-friendly labels, and advertised salary. The analysis is descriptive: it shows patterns in the available dataset and does not establish cause and effect.

2. Dataset and preparation

Source: Indian Job Market Dataset 2025–2026 on Kaggle
Original dataset: 97,929 rows × 17 columns
After duplicate job IDs were removed: 97,679 rows
Key preparation: inspect missing values and data types; remove duplicate `jobId` records; clean text fields; derive role category, primary city, salary midpoint, average experience, seniority, work mode, and fresher-friendly status.

The source data file is not included in this report. The notebook must be pointed to a locally downloaded copy of the dataset before rerunning.

3. Findings at a glance

Fresher-friendly share: The notebook reports approximately 36% in Ahmedabad and 35% in Noida, compared with around 15–16% in Bengaluru, Hyderabad, and Pune. Share should not be confused with total job count.

Work mode: 89,969 postings are reported as on-site, 5,752 hybrid, and 1,958 remote, so the data is strongly weighted toward on-site roles.

Seniority and salary: Reported median salary midpoints range from about 3.0 LPA for Junior to 7.2 LPA for Lead. The Junior salary sample is comparatively small, and job-role mix can influence group results.

Experience: The notebook suggests a positive association between experience and salary midpoint, but individual outcomes vary.

Employer concentration: Companies with 100+ postings account for around 36% of postings, according to the notebook; smaller recruiters account for around 64%.

Figure 1. Fresher-friendly share by selected city.

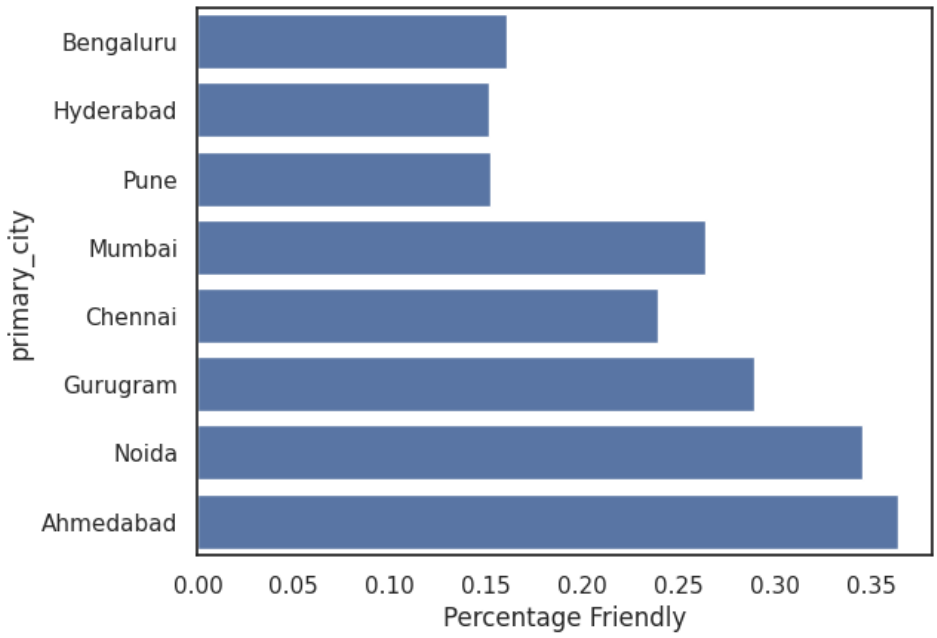


Figure 2. Salary midpoint by seniority.



Figure 3. Salary and experience pattern.

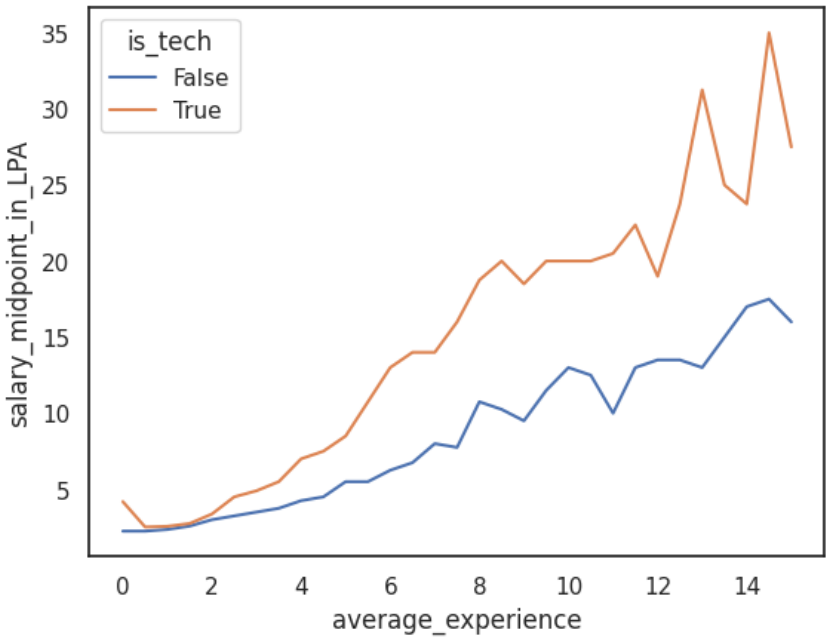
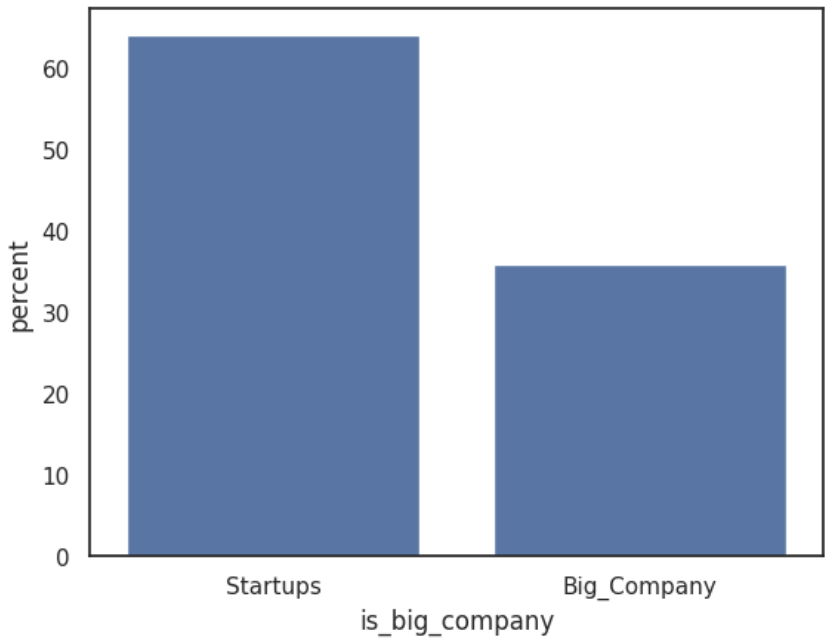


Figure 4. Postings from large and smaller recruiters.


60
50
40
30
20
10
0
percent
Startups
Big_Company
is_big_company

4. Hypothesis testing

The notebook specifies a 95% confidence level ($\alpha = 0.05$). Its reported tests found statistically significant differences/associations for all five questions below.

Test	Question	Reported result
Two-proportion z-test	Fresher-friendly share: tech vs non-tech	6.78% vs 33.06%; $p < 0.05$
Two-sample t-test	Mean disclosed INR salary: tech vs non-tech	■14.53 LPA vs ■6.88 LPA; $p < 0.05$
Chi-square	Work mode vs fresher-friendly status	$\chi^2 \approx 1240.34$, df = 2; $p < 0.05$
One-way ANOVA	Salary midpoint across seniority	$F \approx 142.15$; $p < 0.05$
One-sample proportion z-test	Fresher-friendly share vs 25% benchmark	$z \approx 23.10$; $p < 0.05$

Interpretation: statistical significance is not proof of causation and does not automatically mean a difference is practically large. Notebook p-values displayed as 0.0 should be understood as extremely small values rounded by output formatting, not literal zero.

5. Tools and reproducibility

Python, pandas, NumPy, Matplotlib, Seaborn, SciPy, statsmodels, and Jupyter Notebook/Google Colab. Download the Kaggle dataset, update the path in the notebook, install required packages, and run cells from top to bottom. Some cells use Google Colab/Drive-specific paths.

6. Limitations

Findings depend on dataset coverage and the notebook's derived labels. Salary analysis uses postings with disclosed usable salary values and selected filters. Outliers and group sizes affect comparisons. The notebook contains exploratory/intermediate cells; rerun end-to-end and verify outputs before presenting the results as final.

7. Conclusion

The project demonstrates data cleaning, feature engineering, exploratory visualisation, salary and hiring analysis, and hypothesis testing. For job seekers, compare job volume, fresher-friendly share, and salary as separate indicators rather than relying on one measure alone.